

Linear Algebra (brief review)

- Mathematical objects:

scalar $x \in \mathbb{R}$

vector $\vec{x} \in \mathbb{R}^n$

matrix $X \in \mathbb{R}^{n \times m}$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad x_i \in \mathbb{R} \quad \forall i \in \{1, 2, \dots, n\}$$

$$\begin{bmatrix} \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{bmatrix}_{n \times m} \quad x_{i,j} \in \mathbb{R} \quad \begin{matrix} \forall i \in \{1, 2, \dots, n\} \\ \forall j \in \{1, 2, \dots, m\} \end{matrix}$$

- Vector norms:

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

L0 norm

$$\|\vec{x}\|_0 = \# \text{ non-zero entries in } \vec{x}$$

L1 norm

$$\|\vec{x}\|_1 = |x_1| + |x_2| + \dots + |x_n|$$

L2 norm

$$\|\vec{x}\|_2 = \sqrt{|x_1|^2 + |x_2|^2 + \dots + |x_n|^2}$$

L- ∞ norm

$$\|\vec{x}\|_\infty = \max \{ |x_1|, |x_2|, \dots, |x_n| \}$$

$\xrightarrow{p \geq 1}$
L^p-norm

$$\|\vec{x}\|_p = \sqrt[p]{|x_1|^p + |x_2|^p + \dots + |x_n|^p}$$

Equivalency of norms

$$\alpha_1 \|\vec{x}\|_q \leq \|\vec{x}\|_p \leq \alpha_2 \|\vec{x}\|_q$$

- Inner (dot) product:

$$\vec{x} \in \mathbb{R}^n$$

$$\vec{y} \in \mathbb{R}^n$$

$$\longrightarrow \langle \vec{x}, \vec{y} \rangle = \vec{x}^T \vec{y} = \vec{y}^T \vec{x} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

- Matrix operations:

- product: $X \in \mathbb{R}^{n \times m}$, $Y \in \mathbb{R}^{m \times p}$

$$Z = XY \in \mathbb{R}^{n \times p}$$

$$\begin{bmatrix} \text{---} \end{bmatrix}_{n \times m} \begin{bmatrix} \text{---} \end{bmatrix}_{m \times p} = \begin{bmatrix} \text{---} \end{bmatrix}_{n \times p}$$

- inverse: $X \in \mathbb{R}^{n \times n}$ if invertible $\rightarrow X^{-1} \in \mathbb{R}^{n \times n}$

$$XX^{-1} = X^{-1}X = I_{n \times n}$$

$$I_{n \times n} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix}$$

- transpose: $X \in \mathbb{R}^{n \times m} \rightarrow X^T \in \mathbb{R}^{m \times n}$

$$\begin{bmatrix} \text{---} \end{bmatrix} \rightarrow \begin{bmatrix} \text{---} \end{bmatrix}$$

- eigenvalue and eigenvector: $X \in \mathbb{R}^{n \times n}$

Left eigenvector

$$\vec{v} X = \lambda \vec{v}$$

$$X \vec{v} = \lambda \vec{v}$$

$\uparrow$ eigenvector
 $\uparrow$ eigenvalue

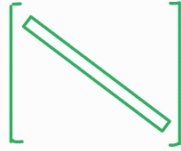
- eigendecomposition: $X \in \mathbb{R}^{n \times n}$

if diagonalizable $X = V \Lambda V^T \rightarrow X = \sum_{i=1}^n \lambda_i v_i v_i^T$

$$\begin{bmatrix} \text{---} \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix}$$

• trace : $X \in \mathbb{R}^{n \times n}$

$$\text{Tr}(X) = x_{11} + x_{22} + \dots + x_{nn}$$



$$\text{Tr}(X) = \sum_{i=1}^n \lambda_i$$

• determinant : $X \in \mathbb{R}^{n \times n}$

$$|X| = \prod_{i=1}^n \lambda_i$$

Probability (brief review)

• Probability space : $(\Omega, \mathcal{F}, P)$

• sample space Ω : set of all possible outcomes

• event space $\mathcal{F}$: set of events, where each event is a subset of sample space Ω

* $\mathcal{F}$ should be a σ -algebra, i.e.,

• $\Omega \in \mathcal{F}$

• $A \in \mathcal{F} \Rightarrow A^c = \Omega \setminus A \in \mathcal{F}$

• if $A_1, A_2, \dots \in \mathcal{F} \Rightarrow A = \bigcup_i A_i \in \mathcal{F}$

De Morgan's Law

$\bigcap_i A_i \in \mathcal{F}$

$(\Omega, \mathcal{F})$ is called a measurable space.

- probability measure (function) P : Assigns a probability (likelihood) value to each event

$$P: \mathcal{F} \longrightarrow [0, 1]$$

Properties of P :

$$\bullet P[\Omega] = 1$$

$$\bullet P[A_1 \cup A_2] = P[A_1] + P[A_2]$$

$$\forall A_1, A_2 \in \mathcal{F} \text{ s.t. } A_1 \cap A_2 = \emptyset$$

Example:

1) Flipping a coin

$$\Omega = \{H, T\}$$

$$\nearrow \mathcal{F} = \{\emptyset, \Omega\} \quad \begin{cases} P[\emptyset] = 0 \\ P[\Omega] = 1 \end{cases}$$

$$\searrow \mathcal{F} = \{\emptyset, \{H\}, \{T\}, \Omega\} \quad \begin{cases} P[H] = 0.5 \\ P[T] = 0.5 \end{cases}, \begin{cases} P[\emptyset] = 0 \\ P[\Omega] = 1 \end{cases}$$

2) Rolling a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\mathcal{F} = 2^\Omega = \{\emptyset, \{1\}, \dots, \{6\}, \{1, 2\}, \dots, \{5, 6\}, \dots, \Omega\}$$

$$\begin{cases} P[1] = \frac{1}{6} \\ P[2] = \frac{1}{6} \\ \vdots \\ P[6] = \frac{1}{6} \end{cases} \longrightarrow P[\{1, 2, 3\}] = \frac{3}{6}$$

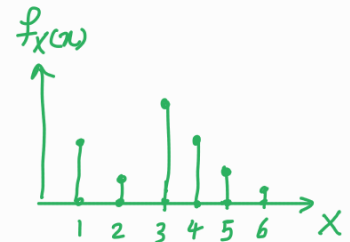
- Random variable: A measurable function/mapping $X: \Omega \rightarrow E$

$$\Omega = \{H, T\} \quad E = \{1, 0\}$$

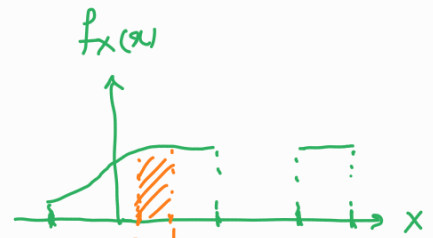
$$X: \{H, T\} \rightarrow \{1, 0\} \quad \begin{cases} X(H) = 1 \\ X(T) = 0 \end{cases}$$

$E = \mathbb{R}$ is typically used with Borel σ -algebra $\mathcal{B}(\mathbb{R})$

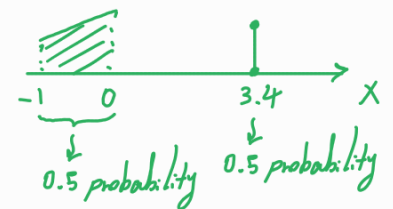
Types $\rightarrow$ Discrete

$$P[X=x] = p_X(x) \quad \text{PMF}$$


$\rightarrow$ Continuous

$$P[a \leq X \leq b] = \int_a^b f_X(x) dx \quad \text{PDF}$$


$\rightarrow$ Mixed



CDF $F_X(x) = P[X \leq x] = P[-\infty < X \leq x]$

- Conditional probability: Let A and B be two events

$$P[A|B] = \frac{P[A \cap B]}{P[B]} \quad \begin{array}{l} \xrightarrow{\text{independent}} P[A]P[B] \\ \xrightarrow{\text{not independent}} P[B]P[A|B] \end{array}$$

> 0 ✓

If $P[A] > 0$ & $P[B] > 0 \rightarrow P[A|B]$ & $P[B|A]$ defined ✓

$$P[A]P[B|A] = P[A \cap B] = P[B]P[A|B] \xrightarrow[\text{Rule}]{\text{Bayes}} P[A|B] = \frac{P[A]P[B|A]}{P[B]}$$

- Expectation:

discrete RV $E[X] = \sum_{x \in X} x f_X(x)$ $\nwarrow P[X=x]$

flipping a coin $\begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$
 $E[X] = p$

rolling a fair die $E[X] = 3.5$

continuous RV $E[X] = \int x f_X(x) dx$

uniform distribution $f_X(x)$ on $[a, b]$
 $E[X] = \frac{a+b}{2}$

normal distribution $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$
 $E[X] = \mu$

- Conditional expectation: Let X and Y be two random variables

$E_X[X|Y]$ $\rightarrow$ a function of Y

$\rightarrow$ a particular value

$\xrightarrow[\text{discrete } Y]{\text{discrete } X} E_X[X | Y=y] = \sum_x x P[X=x | Y=y]$